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We modelled patterns of collaboration, team gender composition, and funding amounts across awarded Australian government-funded competitive team research grants from 2000 to 2020. The percentage of grants awarded to women at mid- and senior-career levels was higher for team grants than sole investigator grants compared to men at those levels. Teams led by women tended to have a greater percentage of women coinvestigators than teams led by men, but this was below gender parity regardless of team leader gender. Funding amounts per grant did not differ by the principal investigator's gender and reached parity in 2020 across teams with both high and low representation of women. Since teams tend to be more gender-balanced when led by women, women's grant leadership may be an important mechanism for shifting overall representation of women in research. We offer public policy measures to address gender inequities in the research sector.

Keywords: gender; grants; research; workforce.

1. Introduction

Despite widespread efforts to promote gender equity in research, women continue to be under-represented, especially at senior levels and in science, technology, engineering, and mathematics (STEM) fields (European Commission Directorate-General for Research and Innovation 2021; UNESCO et al. 2021; Australian Government 2022). A key area of interest in understanding these gender differences is competitive research grant funding (Ceci, Kahn, and Williams 2023). As research becomes increasingly collaborative (West et al. 2013; Leahey 2016), it is crucial to understand the gender dynamics of research teams within competitive grant programmes. Yet such dynamics are currently insufficiently understood.

In a comprehensive analysis of Australian national competitive grants awarded over two decades (2000–20), we (I.K., E.S., L.H.-S., E.L.J., and L.A.W.) explored gender differences in awarded grants relative to application rates and research workforce participation by gender (Kingsley et al. 2023). We found that observed gender differences in awarded research grants mirror unequal participation in the research workforce. Our findings imply that fewer women in the research workforce and leading grant applications likely result in fewer awarded grants led by women than by men. These findings highlight the need for a deeper understanding of gender dynamics in research grant outcomes, particularly in collaborative research contexts.

Building on these insights, the objective of the present study was to investigate the gender dynamics of awarded team

grants in Australia, with a specific focus on patterns of collaboration (team leader versus sole investigator), and team gender composition based on the principal investigator's gender. Our original contribution lies in providing new evidence on these gender dynamics, particularly in the previously unexplored area of collaboration patterns by gender within grant outcomes.

Grant-funded teams generally fall short of gender parity (Bowman and Ulm 2016; Pezzoni and Visentin 2024) and 'gender homophily' effects are documented, wherein men tend to collaborate with other men, while women lead more gender-balanced teams (Bowman and Ulm 2016; Araújo et al. 2017; Pezzoni and Visentin 2024). However, the impact of team gender composition on grant outcomes remains unclear and mixed (Lungeanu, Huang, and Contractor 2014; Wu et al. 2021; Bianchini et al. 2022). Furthermore, no studies have investigated patterns of research collaboration—specifically, working in teams versus working alone—based on gender within grant outcomes, although publication patterns suggest women tend to work in teams more often than do men (Jamali and Abbasi 2023).

To address this gap in the literature, we examined awarded team grants from Australia's two major public research funders: the Australian Research Council (ARC) and the National Health and Medical Research Council (NHMRC). We focused exclusively on grants in schemes conducive to collaboration—that is, those that may be awarded to groups of two or more individuals, although some may also be awarded to single investigators. Our analysis examines

35,777 awarded team grants over the period of 2000–20 using a regression approach that accounts for factors such as the principal investigator's career seniority, team size, field of research, and the research intensity of the host organization.

Our study makes several key contributions to the literature by providing empirical evidence on gender dynamics in awarded team research grants. We provide the first comprehensive analysis of collaboration patterns by gender within team grant outcomes, revealing that the percentage of grants awarded to women at mid- and senior-career levels was higher for team grants than sole investigator grants compared to men at those levels. Additionally, we offer new insights into the gender composition of teams based on the principal investigator's gender, demonstrating that teams led by women had a greater percentage of women than those led by men, although overall teams fell short of gender parity (i.e. 50 per cent of women within a team), regardless of the gender of the team leader. Finally, we analyze funding amounts based on the gender and role of the principal investigator, and on team gender composition. Although funding per grant did not differ by the gender of the principal investigator, cumulative funds awarded to grants led by women were \$7.2 billion lower than those awarded to grants led by men—an outcome driven by numerically fewer awarded grants being led by women.

Our findings reveal a potential avenue to increase the representation of women in research. Because teams tend to be more gender-balanced when led by women, women's grant leadership may be an important mechanism for shifting overall representation of women in grant programmes and beyond. Based on our findings, we discuss policy implications aimed at (1) prioritizing collaborative grants as a means to increase women's participation in both grant leaders and team members, (2) incentivizing gender diversity in teams, and (3) strengthening institutional accountability for gender equity. By providing an empirical foundation for evidence-based interventions, our work underscores women's team leadership as a potentially powerful lever for sustainable change in addressing gender inequities in research.

The remainder of this paper is organized as follows: after this introduction, we summarize prior literature and provide contextual information regarding the funding bodies in [Section 2](#); in [Section 3](#), we explain our methodological approach; and in [Section 4](#), we present the results of our modelling. We offer reflections, policy implications, and conclusions in [Section 5](#).

2. Background

2.1 Mixed and limited evidence on gender dynamics in awarded team grants

Worldwide, gender disparities within the research sector are well documented ([European Commission Directorate-General for Research and Innovation 2021](#)). Despite initiatives to promote gender equity, cultural and systemic challenges continue to hinder women's participation in research, especially in STEM fields and at senior levels of career seniority ([European Commission Directorate-General for Research and Innovation 2021](#); [UNESCO et al. 2021](#); [Australian Government 2022](#)).

The allocation of research grants is of key interest in relation to understanding gender disparities ([Ceci, Kahn, and Williams 2023](#)). Receiving research grants not only supports

research activities ([Wadman 2009](#)) but is also a key success indicator for career advancement ([Liner and Sewell 2009](#); [Bonetta 2011](#)). While gender differences have been identified in competitive grant programme outcomes, the evidence is mixed, and the nature and source of these differences remain unclear. Some evidence suggests that the grant application assessment process is prone to biases against women ([Pohlhaus et al. 2011](#); [Bedi, Van Dam, and Munafo 2012](#); [Hechtman et al. 2018](#); [Spekkens, Cofie, and Crabtree 2018](#); [Tamblyn et al. 2018](#); [Oliveira et al. 2019](#); [Witteman et al. 2019](#); [Safdar et al. 2021](#); [Suarez, Fiorentin, and Pereira 2023](#)). Conversely, two meta-analyses reveal no evidence of gender bias in peer-reviewed grant outcomes ([Ceci, Kahn, and Williams 2023](#); [Schmaling and Gallo 2023](#)), aligning with the premise that gender differences in grant outcomes may primarily arise from unequal workforce participation ([Boyle et al. 2015](#); [Kingsley et al. 2023](#)). The ambiguity in the literature is a key gap that our study aims to address.

As research becomes increasingly collaborative ([West et al. 2013](#); [Leahey 2016](#)), patterns of research collaboration within grant programmes according to gender—specifically, working in teams versus working alone—deserve investigation. Adjacent research on academic publication authorship reveals that women publish more coauthored than sole-authored articles compared to men ([West et al. 2013](#); [Jamali and Abbasi 2023](#)). The percentage of women sole-authors has increased over time but remains below 50 per cent ([West et al. 2013](#)). Our study extends this line of inquiry into the realm of research grants, examining whether similar collaboration patterns exist among principal investigators of awarded team research grants.

Extending on collaboration, evidence suggests that grant-funded research teams fall short of gender parity. Regardless of the gender of the team leader, research grant teams have less than 50 per cent women on average ([Bowman and Ulm 2016](#); [Pezzoni and Visentin 2024](#)). Lower proportions of women in teams may reflect gender bias, or a scarcity of women researchers to include on teams ([Card et al. 2023](#)), while still below parity, teams led by women tend to have a higher percentage of women investigators compared to those led by men ([Bowman and Ulm 2016](#); [Pezzoni and Visentin 2024](#)). For example, recent research found gender bias against women in grant team formation, with a weaker pattern among teams led by women and by early-career researchers ([Pezzoni and Visentin 2024](#)). Our study builds on these findings by examining team gender composition across a large dataset of awarded grants over two decades, providing insights into the extent of gender disparity in grant-funded research teams and how it relates to team leader gender.

'Gender homophily' captures the pattern wherein individuals associate more with others of the same gender ([McPherson, Smith-Lovin, and Cook 2003](#)). Across research more broadly, gender homophily arises in various domains, including coauthorship ([Kwiek and Roszka 2021](#)), hiring of research staff ([Sheltzer and Smith 2014](#); [Kewley et al. 2023](#)), and research supervision ([Kewley et al. 2023](#); [Mühleck and Schwabe 2024](#)). Gender homophily is stronger among men than women ([Kwiek and Roszka 2021](#)), while women are more egalitarian than men ([Araújo et al. 2017](#)). Intriguingly, gender homophily could present a mechanism for increasing the representation

of women in research. For instance, when women PhD candidates work with women supervisors, they are more likely to continue in academia (Gaule and Piacentini 2018). Similarly, when women lead research teams, gender homophily can help attract and retain other women researchers (Kewley *et al.* 2023). Our study examines whether these patterns of gender homophily extend to grant-funded research teams and explores their implications for gender equity in research.

Critically, the evidence regarding the influence of team gender composition on grant outcomes is mixed. While some evidence suggests that research teams with at least one woman have higher success rates than teams with no women (Lungeanu, Huang, and Contractor 2014), other evidence indicates that mixed-gender teams are awarded fewer grants than same-gender teams (Wu *et al.* 2021). Some research reports that grant applications from teams with a higher proportion of women receive less favourable evaluations (Bianchini *et al.* 2022; Foss and Olsen 2023), although this effect weakens with increasing project heterogeneity (in terms of industry affiliations) and novelty (Foss and Olsen 2023). One study revealed a consistent disconnect between reviewers' comments and scores. Namely, reviewers' comments did not indicate that they perceived women researchers as being less competent, but applications with more women were scored less positively (Bianchini *et al.* 2022). In relation to funding amounts, some evidence suggests that women-only research teams attract less funding than men-only teams, and mixed-gender teams attract more funding than same-gender teams (Wu *et al.* 2021). Our study contributes to this debate by examining the relationship between team gender composition and grant funding amounts, offering new insights into the influence of team gender composition on grant outcomes.

We acknowledge that people with nonbinary gender identities face barriers to participation in mixed-gender teams (Lansu, Bleijenberg, and Benschop 2019; Marin-Spiotta *et al.* 2023). Our dataset, however, did not allow for the examination of nonbinary gender identities. Acknowledging the limitations of binary approaches to gender, any research that advances understanding of the complex, intersecting factors that prevent or promote gender equity in research holds great value.

2.2 Recent findings on Australian research grant awards, applications, and workforce participation

A comprehensive analysis of Australian national competitive grants awarded from 2000 to 2020 provides an important context for understanding gender dynamics in Australian research (Kingsley *et al.* 2023). The study examined 46,912 grants awarded to 17,130 unique lead investigators by the ARC and the NHMRC, incorporating both grant application and research workforce data. The study revealed fewer awarded grants were led by women than men; however, overall success rates of grant applications did not vary according to lead investigator gender. There were fewer women than men in the research workforce. The award rate (awarded grants relative to workforce participation) was slightly higher for women than men. Most of these observed gender differences were largest at senior-career levels. Together, these patterns imply that fewer women in the research workforce and leading grant applications have resulted in fewer awarded grants led by women than by men. This pattern emerges despite a disparity in award rates favouring women. Our current study

extends these insights by focusing exclusively on awarded grants in schemes conducive to collaboration, examining patterns of collaboration and team gender composition based on the principal investigator's gender.

2.3 The ARC and the NHMRC

Our present study examines collaborative research grants awarded by the two primary Australian government funding agencies from 2000 to 2020: the ARC and NHMRC.

The ARC is the Australian Government's main funding agency for nonmedical research. Across the period covered by this study, the ARC awarded a total of \$12.9 billion in funding to support basic and applied research, research training programmes, collaborative research initiatives, and infrastructure projects (Australian Research Council 2022a). The ARC's competitive-peer reviewed grants are distributed through two main programmes: the Discovery Program and the Linkage Program. The Discovery Program supports individuals and small teams through schemes such as the Discovery Projects, Discovery Early Career Researcher Award, Future Fellowships, Australian Laureate Fellowships, and Discovery Indigenous. The Linkage Program, which fosters university-industry collaborations, contains ten schemes such as Centres of Excellence, Industrial Transformation Research Program, Linkage Infrastructure, Early Career and Mid-Career Industry Fellowships, and Industry Laureate Fellowships (Australian Research Council 2022a).

The NHMRC is the Australian Government's principal research funding agency for health and medical research. Across the period of this study, the NHMRC awarded a total of \$13.5 billion in funding (National Health and Medical Research Council 2021). Following a 2018–9 reform, NHMRC offers four competitive, peer-reviewed grant schemes: Investigator Grants (providing salary and research support), Ideas Grants (for innovative projects), Synergy Grants (for multi-disciplinary teams addressing complex questions), and Strategic and Leveraging Grants (targeting national research priorities) (National Health and Medical Research Council 2021).

Over the period of 2000–20, both the ARC and NHMRC made several changes to their funding schemes to better align with national research priorities. These changes include the introduction of new grant schemes, as well as modifications to and discontinuation of existing ones, reflecting emerging research priorities and policy directions (Australian Government 2024).

While the structure of their funding schemes has evolved, the core elements of their peer review processes have remained relatively consistent. Both the ARC and NHMRC employ a multi-stage, single-blind peer review process where reviewers know the identity of the applicants, but not vice versa (National Health and Medical Research Council 2022; Australian Research Council 2022b). This process involves external assessors and advisory panels that review the assessors' evaluations and make funding recommendations to the relevant Australian Government Ministries.

To address discipline-specific needs, both agencies have assessors and/or panels with expertise in relevant fields of research. Assessors evaluate applications against specific criteria using a standardized scoring system and provide written feedback. Applications are assessed based on the quality of the research proposal, the investigator's track record, and the

potential impact of the research, although specific application guidelines and assessment criteria may vary between schemes (National Health and Medical Research Council 2022; Australian Research Council 2022b). A notable change to the ARC assessment process within the period covered by the study was the introduction of a criterion in 2011 that enables evaluation of a researcher's activities, outputs, and achievements (i.e. track record) in the context of career and life opportunities and experiences, including career interruptions (Australian Research Council 2023).

2.4 Aim and research questions

While existing literature and recent comprehensive Australian analysis of awarded grants provide valuable insights into gender dynamics in research, substantial gaps remain, particularly regarding the specifics of collaboration patterns and team gender composition. Our research aimed to address these gaps by examining the gender dynamics of awarded team grants in Australian government-funded competitive research grant schemes. Building on the broader context of gender disparities in Australian grant awards, applications, and workforce participation, we focused specifically on collaborative grants to provide a more nuanced understanding of gender dynamics.

Specifically, we applied an observational approach to examine the patterns of collaboration and team gender composition based on the gender of the principal investigator and to explore funding amounts based on the principal investigator's role and the gender composition of teams. The data records from both funding bodies only included data on successful grant applications; therefore, our results exclusively relate to awarded grants and do not reflect patterns among applications or success rates. However, we argue that these data are appropriate as they shed light on research sector outcomes that could be driven by biases in the grant review process, extending the comprehensive analysis by Kingsley et al. (2023). Our research questions were the following:

RQ1: Do academic level and being a sole investigator (versus team leader) predict the proportion of grants awarded to women relative to men?

RQ2: Among grants awarded to teams of investigators, do the gender and academic level of the team leader predict the gender composition of the other investigators within the team?

RQ3: Do sole investigator versus team leader status, the principal investigator's gender, and the principal investigator's academic level predict grant funding amount?

RQ4: Among grants awarded to teams of investigators, do team gender composition (including the team leader) and the size of the team predict grant funding amount?

Research questions 1–4 also assess trends over time, as well as whether funding body (ARC/NHMRC), field of research, and the research intensity of the host organization predict the relevant outcome.

3. Methods

3.1 Data collection and processing procedures

We used existing data records of grants awarded by the ARC and NHMRC. The overall dataset contained 46,912 awarded grants to 17,130 unique lead investigators over two decades (2000–20). Most ARC data comprised public records published by the ARC on their website, with additional information provided by the ARC upon request. All NHMRC data were provided by the NHMRC. Together, the records yield a 21-year dataset for both funding bodies.

The awarded grants dataset did not include investigators' self-identified gender. We requested these data at the application level, but both funding bodies declined, citing privacy reasons. Consequently, we interpolated the likely gender of each principal investigator as 'woman' or 'man'¹ based on their first name using the *genderize.io* (Genderize.io, n.d.) application programming interface (API) via the R package *DemografixeR* (Brenninkmeijer 2020). We acknowledge the limitations of this method, including the binarization of a nonbinary construct and risks of misgendering investigators (Santamaría and Mihaljević 2018). However, we argue that this method is suitable for the purposes of this research, which examines potential disparities stemming from the *perceived* gender of an investigator by grant application assessors (Van der Lee, Ellemers, and Fiske 2015; Witteman et al. 2019). Our method does not exclude nonbinary investigators but instead captures potential bias against them due to the likely assumption of *binary* gender by grant application assessors (Shelton and Dodd 2021).

We applied the 'genderize' function within *DemografixeR* to the extracted first names of all investigators. The function provides one of three gender classification responses for each first name: male, female, or none. In this article, we use the term 'women' to represent individuals whose first names were classified as female and 'men' to represent those classified as male. The API also provides a probability score, which indicates the certainty of the classified gender. At the time of analysis, the API's underlying dataset contained 114,451,298 entries of more than 250,000 names across 242 international standard country codes. The service provider does not declare how data were gathered, nor sources used; therefore, its reliability is difficult to evaluate (Blevins, Blevins, and Mullen 2015; Santamaría and Mihaljević 2018). While *genderize.io* achieves high accuracy with data from many countries (VanHelene et al. 2024) (i.e. Germany, 98.9 per cent; Japan, 98.2 per cent; France, 97.2 per cent; and the USA, 96.6 per cent), its performance is slightly lower for data from Australia (86 per cent) (Jadidi et al. 2018) and notably lower with data from South Korea, China, Singapore, and Taiwan (<82 per cent) (VanHelene et al. 2024). As noted later, in our modelling we weighted the uncertainty of the predicted gender using the probability score. The *genderize.io* API has been widely used in gender-related analyses in areas such as peer review (Fox, Burns, and Meyer 2016; Fox and Paine 2019; Kiang et al. 2022), citation impact of scientific publications by author teams of different gender compositions (Kuchanskyy et al. 2023), and authorship in journals of various fields, including medicine (DeFilippis et al. 2021; Hart and Perlis 2021; Gadek et al. 2022; Batumalai et al. 2023) and seismology (Ermert, Koroni, and Korta Martiartu 2023).

We engaged in data processing as per the plan detailed at <https://osf.io/2fr9t/>. For the present analysis, which builds on

the comprehensive analysis of ARC- and NHMRC-awarded grants (Kingsley *et al.* 2023), we excluded 10,187 grants from schemes that solely awarded grants to individual recipients and did not allow for teams of two or more (e.g. fellowships). This exclusion enabled us to focus our study on grants conducive to collaboration—that is, those allowing groups of two or more individuals, although some may also be awarded to single investigators in order to examine patterns of collaboration, team gender composition, and funding amounts. In the remaining dataset of grants allowing two or more individuals, we created a three-factor categorical variable to represent team size [i.e. team size (categorical)], which includes the principal investigator: ‘single’ for grants in schemes allowing groups of two or more individuals but only included a single principal investigator without a team (i.e. sole investigator), ‘small’ for teams with two to four investigators, and ‘large’ for teams with five or more investigators.

The final dataset of team grants, including inferred principal investigator gender; role; actual, inferred, and imputed academic level of principal investigators; team size (categorical); team gender composition; field of research of awarded grant; and research intensity of the host organization, contained 35,777 awarded grants that were submitted as applications between 2000 and 2020 (ARC=23,759, NHMRC=12,018).

Descriptive statistics of the dataset (i.e. counts and raw percentages of awarded grants, median amount per grant, and total amount of funding over time in billions) by gender, role, academic level, select years, and field of research are provided in [Supplementary Table S1](#).

3.2 Preregistration

We preregistered our research at <https://osf.io/2fr9t/> after initial data processing but before formal statistical analysis. We report the results of four preregistered models in this article (Models 1–4), with procedural modifications detailed in an addendum, also available at the above link. The results of the other models outlined in the preregistration are reported in Kingsley *et al.* (2023).

3.3 Statistical analyses

We performed four regression models (Models 1–4), one for each research question (RQ1–RQ4). *A priori*, we classified focus variables for each research question, but additionally considered secondary variables that might explain variation in the response variable for that question. Statistical analyses consisted of two processes for each model: first, selecting secondary variables for inclusion (see [Section 3.3.2](#)) and second, estimating confidence intervals (CIs) for the effects in the model (see [Section 3.3.3](#)).

3.3.1 Model descriptions The model for awarded grants (Model 1) was a binomial generalized linear mixed model with the gender of the principal investigator as the binary response variable, weighted for uncertainty on inferred gender. Model 1 investigated how the principal investigator’s academic level and being a sole investigator (versus a team leader) predict the proportion of grants being awarded to women (RQ1). We excluded schemes that exclusively award grants to groups of two or more individuals, as collaboration is mandatory for these schemes, and therefore, they do not involve sole investigators (e.g. ARC Centres of Excellence

and NHMRC Partnership Grants). Focus variables were time, principal investigator academic level, and role (and two-way interactions between them).

The model for team gender composition (Model 2) was a generalized linear mixed model with the proportion of women within teams as the Poisson response variable, offset on team size excluding the principal investigator (log transformed). Model 2 investigated how team gender composition (proportion of women:men collaborators) is predicted by the principal investigator’s gender and academic level (RQ2). We excluded grants awarded to sole investigators due to the absence of other investigators on these grants. Focus variables included time, principal investigator gender, and academic level (and two-way interactions between them).

The models for funding amount (Models 3 and 4) were both linear mixed models with a continuous response variable for awarded funding amount (as a proportion of the maximum amount awarded per scheme per year). Model 3 investigated how awarded funding amount is predicted by whether the principal investigator is a sole investigator versus team leader, their gender, and their academic level (RQ3) using the data subset used in Model 1. Focus variables were time, principal investigator gender, academic level, and role (and two-way interactions between them). Model 4 investigated how team gender composition and team size predicted funding amount (RQ4) using the data subset used in Model 2. The focus variables for Model 4 were time, team size (categorical), and team gender composition (proportion of women:men investigators including the principal investigator)—and two-way interactions between them. Per-grant funding amounts were standardized as a proportional deviation from the median amount awarded per scheme per year. Values greater (less) than 1 represent receiving more (less) than the median funding amount within that scheme and year. We log-transformed standardized per-grant funding amounts to obtain a symmetrically distributed variable.

Field of research, scheme, funding body, research intensity of the host organization, and the interaction between each of these and the application year (both linear and quadratic) were classified as secondary variables and considered for inclusion in all four models.

3.3.2 Secondary variable selection We used a data-splitting approach to select secondary variables with one subset of data and then implemented selected models on a second independent subset (Romano and DiCiccio 2019) to enable valid conditional inference. We divided the data into two equal-sized independent ‘splits’, sampling principal investigator IDs to ensure independence and stratifying on principal investigator gender, year, and funding body of their first project to get more homogeneous splits (Reitermanová 2010).

We used a group lasso to determine which secondary variables may explain variation in the response variables (Breheny and Huang 2015). The group lasso, implemented with *grpreg* (Breheny and Huang 2015) and the minimax concave penalty (Reitermanová 2010), allows focus variables to be automatically retained in the variable selection procedure without being penalized and groups of variables to be penalized together (e.g. interactions and main effects, or categorical variable dummy predictors). We ran the group lasso multiple times on each of the thirty imputation iterations of the awarded grants dataset and then selected variables for the

inference set based on majority rule—that is, a variable is selected for inclusion in the models if it is selected in over half the imputation iterations. In each model, a consistent set of secondary variables were selected across imputation iterations. Field of research was selected for all models, research intensity of the host organization was selected for Models 2–4, and scheme was selected for Model 4.² We then assessed whether the residuals required temporal autocorrelation structures by plotting a variogram and checked distributional mean-variance assumptions with residuals versus fitted plots. Neither autocorrelation structures nor adjustments due to unmet assumptions were required.

3.3.3 Inference on effects We next fitted selected models on the inference set using the glmmTMB package (Brooks et al. 2017). Models 1–4 were fitted to each of the thirty imputation iterations using the mitml package (Grund, Robitzsch, and Luedtke 2023) and likelihood ratio test results were pooled using the ‘D4’ method (Chan and Meng 2017) where possible and the ‘D1’ method in settings where ‘D4’ failed to converge. We used Rubin’s rule to pool model coefficients (referred to as $\hat{\beta}$ ’s in the results), contrasts (e.g. comparisons between women and men), and marginal means (e.g. average percentage of grants awarded to women and men) that we report in Section 4. Where selected, scheme and field of research were included as random effects due to the large number of levels these factors could take. Table 1 presents an overview of the statistical modelling approach described later, detailing model type, response variable, focus and secondary variables, random effects, and data subset used for each model.

We estimated 95 per cent CIs of the response variables through time using estimated marginal means (Lenth et al. 2023). Marginal means represent the mean of the response variable (e.g. gender of the principal investigator) for each level of explanatory variable (e.g. principal investigator academic level and time), averaging across levels of other variables. In cases when CIs included the variance of the random slopes for gender, we used the Hessian matrix for parameter estimation (with code based on Mason et al. 2023).

3.4 Ethical considerations

All procedures undertaken in this research were approved by the UNSW Sydney Human Research Ethics Committee (approval number HC3468). Data processing, analysis, and visualization were carried out in R (version 4.3.0).

4. Results

We present results for research questions (RQ1–RQ4) in detail later and summarize the key findings by research question in Table 2. We report odds ratios (ORs) as probability measures for main effects, where a ratio greater than 1 indicates an increase and a ratio less than 1 a decrease in the probability of an event occurring. Ratios indicate evidence for a difference (e.g. between women and men, over time) where CI limits do not overlap 1, with these ratios highlighted in bold. Likelihood ratio tests were conducted to assess the significance of interaction effects and are reported as *F*-statistics. The beta coefficient ($\hat{\beta}$) represents the significance of the effect (main or interaction effect) of time. Asterisks denote levels of significance, where a *P*-value less than 0.05 was considered evidence for the interaction effects: **P* < 0.05, ***P* < 0.01,

****P* < 0.001, and ns indicates nonsignificant effects. Raw descriptive data for awarded grants and funding amounts by principal investigator gender, role, and team gender composition are provided in Supplementary Table S1.

In reporting results, we focus on effects that relate to gender (i.e. principal investigator gender and team gender composition) in this article. Full model results, including effects that do not relate to gender (e.g. time, career seniority, and research intensity of the host organization), are provided in the Supplementary material.

4.1 Awarded grants led by women relative to men by role and academic level, over time (RQ1)

Proportionally fewer grants were awarded to women relative to men for both team grants and sole investigator grants. Differences in the percentages of grants awarded to women relative to men for team and sole investigator grants varied according to career seniority, *F* = 6.233***, as described later.

Among doctoral-level researchers (i.e. Levels A–C), the percentage of grants awarded to women relative to men did not differ between team grants and sole investigator grants [team grants = 33 per cent to women; sole investigator grants = 36 per cent to women (ns)] (Fig. 1). However, among mid- and senior-career researchers, the percentage of grants awarded to women relative to men was higher for team grants than sole investigator grants. Specifically, the percentage of grants awarded to women relative to men was 28 per cent for team grants and 23 per cent for sole investigator grants among associate professors (Level D) and was 21 per cent for team grants and 17 per cent for sole investigator grants among professors (Level E). Across all career levels, the percentage of team and sole investigator grants awarded to women was below gender parity (50 per cent). This pattern aligns with the findings from our broader analysis (Kingsley et al. 2023) that fewer awarded grants were led by women than men and that most observed gender differences were the largest at senior-career levels, mirroring research workforce participation.

The percentage of awarded grants to women for both team and sole investigator grants varied by field of research, ranging from an average of 13 per cent to 53 per cent (Fig. 2). The largest gender disparities against women emerged in the fields of earth sciences (13 per cent of awards to women), chemical sciences (13 per cent), physical sciences (14 per cent), mathematical sciences (14 per cent), and technology (15 per cent). Five fields of research were consistent with gender parity, having percentages of awarded grants to women overlapping 50 per cent in their CIs (i.e. history and archaeology; language, communication, and culture; education; studies in creative arts and writing; and law and legal studies). These patterns did not differ between sole investigators and team leaders (i.e. role was not found to be a significant source of variation with field of research in the secondary variable selection procedure) and mirrored the findings from the comprehensive analysis reported in Kingsley et al. (2023) that gender differences in awarded grants reflect field-specific gender differences in research workforce participation.

The odds of grants being awarded to women relative to men increased over the period of the study (2000–20), $\hat{\beta}$ = 0.04***, when averaged across the academic level and research intensity of the host organization. This temporal trend did not differ according to the principal investigator’s

Table 1. Statistical approach overview for the four regression models.

Statistical model	Response variable	Focus variables	Secondary variables	Random effects	Data subset
Model 1: Do academic level and being a sole investigator (versus team leader) predict the proportion of grants being awarded to women relative to men? (RQ1)					
Generalized linear mixed model Family = binomial	Gender of the principal investigator (binary response variable)	Time, academic level, role (and two-way interactions)	NA	Field of research	Excluded schemes exclusively awarding grants to groups of two or more individuals
Model 2: Among grants awarded to team projects, do the gender and academic level of the team leader predict the gender composition of the collaborators within the team? (RQ2)					
Generalized linear mixed model Family = Poisson	Proportion of women within teams (Poisson response variable) Offset = team size excluding the principal investigator (log transformed)	Time, gender, academic level (and two-way interactions)	Research intensity of the host organization	Field of research	Excluded grants awarded to sole investigators
Model 3: Do being a sole investigator (versus team leader), the principal investigator's gender, and the principal investigator's academic level predict grant funding amount? (RQ3)					
Linear mixed model	Awarded funding amount (continuous response variable) Offset = median funding amount (log transformed)	Time, gender, academic level, role (and two-way interactions)	Research intensity of the host organization	Field of research	Excluded schemes exclusively awarding grants to groups of two or more individuals (same subset as Model 1)
Model 4: Among grants awarded to team projects, do team gender composition (including the team leader) and the size of the team predict grant funding amount? (RQ4)					
Linear mixed model	Awarded funding amount (continuous response variable) Offset = median funding amount (log transformed)	Time, team size (categorical), team gender composition (and two-way interactions)	Research intensity of the host organization	Field of research and scheme	Excluded grants awarded to sole investigators (same subset as Model 2)

An overview of the statistical modelling approach for each of the regression models examining the four research questions. The variable *time* refers to the application year of awarded grants. The variables *gender*, *academic level*, and *role* relate to principal investigators only. The response variable *proportion of women within teams* excludes the principal investigator. The variable *team gender composition* refers to the proportion of women on teams including the principal investigator. Secondary variables represent the variable(s) identified by the model selection process (described in Section 3.3.2) as being significant predictors of the outcome/response variable on a subset of the data. This process yielded a different set of predictors for each model. Note that funding body (ARC versus NHMRC) was not a significant unique predictor in any model.

Table 2. Summary of the results for each of the four research questions.

Description of main finding	Results
RQ1: Do academic level and being a sole investigator (versus team leader) predict the proportion of grants awarded to women relative to men? Differences in the percentages of grants awarded to women relative to men for team and sole investigator grants varied according to career seniority. Among associate professors and professors, the percentage of grants awarded to women relative to men was higher for team grants than sole investigator grants. Regardless of their role as team leader versus sole investigator, women were awarded proportionally fewer grants than men. The percentage of awarded grants to women for both team and sole investigator grants varied by FoR. Temporal trend: the odds of grants being awarded to women relative to men increased over time (2000–20). This trend did not differ according to the principal investigator's role, nor did it differ according to their academic level.	Interaction effect (Role $\times$ Academic level): $F = 6.233^{**}$ Model estimates: percentage of grants awarded to women relative to men: Doctoral-level: sole = 36% (CI 29, 43); team = 33% (CI 27, 40), OR = 0.89 (CI 0.77, 1.02) Associate professors: sole = 23% (CI 17, 30); team = 28% (CI 23, 34), OR = 1.30 (CI 1.02, 1.66) Professors: sole = 17% (CI 13, 22); team = 21% (CI 16, 26), OR = 1.26 (CI 1.07, 1.5) Random effect (FoR): ranging from 13% (CI 9, 17) to 53% (CI 43, 63) of awarded grants led by women Main effect (time): $\hat{\beta} = 0.04^{**}$ Interaction effect (Time $\times$ Role): $\hat{\beta} = 0.0002$ (ns) Interaction effect (Time $\times$ Academic level): $F = 0.567$ (ns)
RQ2: Among grants awarded to teams of investigators, do the gender and academic level of the team leader predict the gender composition of the other investigators within the team? Teams led by women had a greater percentage of women within their teams than teams led by men. Differences in the percentage of women within teams did not differ according to the principal investigator's academic level. Differences in the percentage of women within teams varied by FoR. There was a temporal trend towards gender parity within teams over the period of 2000–20 that differed according to the principal investigator's academic level. Over time, the percentage of women increased within teams led by principal investigators at all levels of career seniority, especially among teams led by doctoral-level researchers and professors.	Model estimates: percentage of women within teams by gender: women = 32% (CI: 28, 38); men = 24% (CI: 20, 28), OR = 1.36 (CI 1.3, 1.41) Interaction effect (Gender $\times$ Academic level): $F = 0.23$ (ns) Random effect (FoR): ranging from 14% (CI 12, 17) to 49% (CI 40, 60) women within teams Interaction effect (Time $\times$ Academic level): $F = 2.645^*$ Model estimates: percentage of women within teams by gender: Doctoral: 2000 = 22% (CI 18, 26); 2020 = 32% (CI 27, 38) Associate professors: 2000 = 26% (CI 22, 32); 2020 = 30% (CI 25, 35) Professors: 2000 = 25% (CI 21, 30); 2020 = 32% (CI 28, 38)
RQ3: Do sole investigator versus team leader status, the principal investigator's gender, and the principal investigator's academic level predict grant funding amount? Funding amounts per grant did not differ according to the principal investigator's gender. The effects of role and career seniority of the principal investigator did not vary significantly by gender. Grants awarded to women team leaders received the same amount of funding per grant as men team leaders, relative to the scheme annual median, as did grants awarded to women and men sole investigators. There was a temporal trend towards funding parity across the principal investigator's gender over the period of 2000–20.	Interaction effect (Gender $\times$ Role): $F = 1.435$ (ns) Interaction effect (Gender $\times$ Academic level): $F = 1.051$ (ns) Model estimates: mean % of the yearly scheme median by role and gender: Sole investigator: women = 80% (CI 74, 87); men = 82% (CI 76, 89), OR = 0.98 (CI 0.94, 1.02) Team leader: women = 92% (CI 85, 99); men = 92% (CI 86, 99), OR = 1.00 (CI 0.98, 1.02) Interaction effect (Gender $\times$ Time): $\hat{\beta} = -0.01^{**}$
RQ4: Among grants awarded to teams of investigators, do team gender composition (including the team leader) and the size of the team predict grant funding amount? Temporal trend towards funding parity over the period of 2000–20 according to the gender composition (including the principal investigator). In 2020, funding parity was reached across teams with both low and high representation of women, marking a shift from historical trends whereby teams with mostly women ($\geq 75\%$ women) received less funding per grant than those with mostly men ($\leq 25\%$ women).	Interaction effect (Team gender composition $\times$ Time): $\hat{\beta} = 0.01^{**}$ Model estimates: mean % of the yearly scheme median by year and team composition: 2000: $\leq 25\%$ women = 100% (CI 91, 109); $\geq 75\%$ women = 93% (CI 84, 102) 2020: $\leq 25\%$ women = 85% (CI 77, 93); $\geq 75\%$ women = 85% (CI 77, 94)

A summary of the results for each of the four research questions. *Time* refers to the application year of awarded grants. *Gender*, *academic level*, and *role* relate to principal investigators only. *Team gender composition* refers to the proportion of women on teams including the principal investigator. *F*-statistics represent the significance of interaction effects. The beta coefficient ($\hat{\beta}$) represents the significance of the effect (main or interaction effect) of time.

Asterisks denote levels of significance, where a *P*-value less than 0.05 was considered evidence for the interaction effects: $*P < 0.05$, $**P < 0.001$, and ns indicates nonsignificant effects. Model estimates provide estimated percentages, mean values, and/or ORs that contextualize the reported effects, with CI representing 95 per cent CI limits. ORs represent probability measures for main effects, where a ratio greater than 1 indicates an increase and a ratio less than 1 a decrease in the probability of an event occurring. Ratios indicate evidence for a difference (e.g. between women and men, or over time) where CI limits do not overlap 1, with these ratios highlighted in bold. For field of research (FoR), included in each model as a random effect, the ranges indicate evidence of variability where CIs of the low and high values do not overlap.

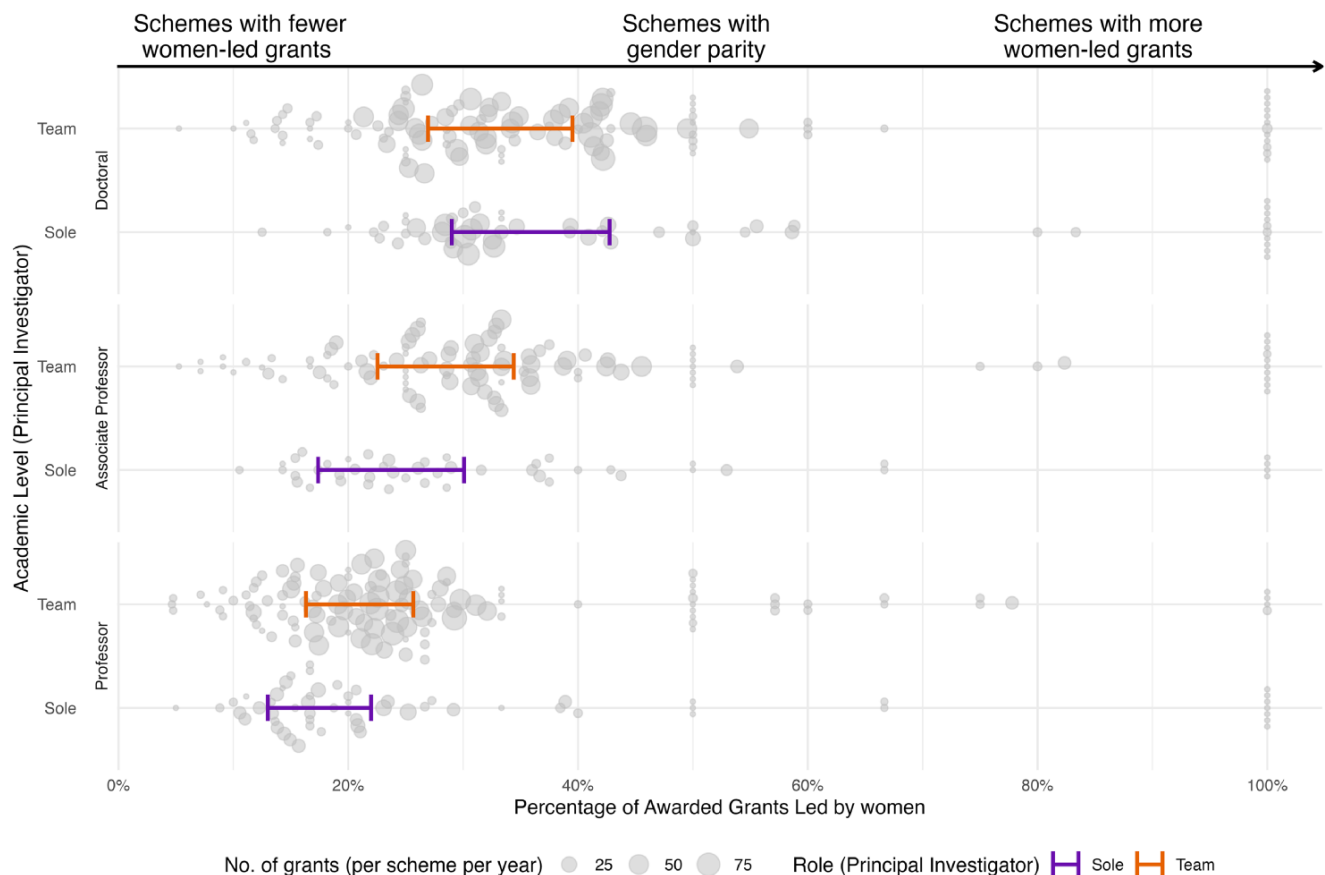


Figure 1. Percentage of awarded grants led by women by role and academic level (RQ1). The percentage of awarded research grants led by women (x-axis) was higher when serving as a team leader compared to being a sole investigator; however, this pattern varied by the academic level of the principal investigator. The horizontal lines represent estimated marginal means, and the vertically widest points of the lines represent the upper and lower CI limits of those estimated means (95 per cent CI). Each dot represents the percentage of awarded grants received by women per a scheme per year, in each academic level and role. The size of the dots is proportional to the number of awarded grants and dots are arranged in a beeswarm (Clarke, Sherrill-Mix, and Dawson 2017) to minimize overlap. Among doctoral-level researchers, 33 per cent (CI 27, 40) of grants awarded to team leaders went to women, while 36 per cent (CI 29, 43) of grants awarded to sole investigators went to women, OR = 0.89 (0.77, 1.02). Among associate professors, 28 per cent (CI 23, 34) of grants awarded to team leaders went to women, while 23 per cent (CI 17, 30) of grants awarded to sole investigators went to women, OR = 1.30 (1.02, 1.66). Among professors, 21 per cent (CI 16, 26) of grants awarded to team leaders went to women, while 17 per cent (CI 13, 22) of grants awarded to sole investigators went to women, OR = 1.26 (1.07, 1.50). Overall, the percentage of grants led by women (relative to men) when serving as team leaders and sole investigators was well below gender parity (50 per cent) across all academic levels. [Supplementary Figure S1](#) illustrates the percentage of awarded research grants led by women for NHMRC-awarded grants from 2000 to 2009 for which the academic title of the principal investigator was missing.

role, $\hat{\beta} = 0.0002$ (ns), nor did it differ according to the principal investigator's academic level, $F = 0.567$ (ns). Again, this pattern mirrored the findings from the comprehensive analysis of the full dataset reported in Kingsley *et al.* (2023).

4.2 Team gender composition by the gender and academic level of the team leader, over time (RQ2)

Overall, team gender composition of grants awarded to teams of investigators fell well short of gender parity (i.e. 50 per cent of women within a team), regardless of the gender of the team leader. Teams led by women had a greater percentage of women than teams led by men. Specifically, teams led by women had 1.36 (95 per cent CI 1.40, 1.41) times more women within their teams compared to teams led by men, averaging across time, academic level, and research intensity of the host organization. Teams led by women comprised an estimated average of 32 per cent women, and teams led by men comprised 24 per cent women (Fig. 3). These patterns

did not significantly differ according to the career seniority of the principal investigator, $F = 0.23$ (ns).

We observed a temporal trend towards gender parity in team composition over the period of 2000–20 that differed according to the academic level of the principal investigator, $F = 2.645^*$. Over time, the percentage of women increased within teams led by principal investigators at all levels of career seniority, especially among teams led by doctoral-level researchers and professors. The percentage of women increased from 22 per cent in 2000 to 32 per cent in 2020 among teams led by doctoral-level researchers, from 26 per cent in 2000 to 30 per cent in 2020 among teams led by associate professors, and from 25 per cent in 2000 to 32 per cent in 2020 among teams led by professors. However, these percentages still remained well below parity in 2020.

Team gender composition varied by field of research, ranging from an average percentage of 14 per cent to 49 per cent women within teams (Fig. 2). Women were least represented among teams in the fields of physical sciences (14 per cent),

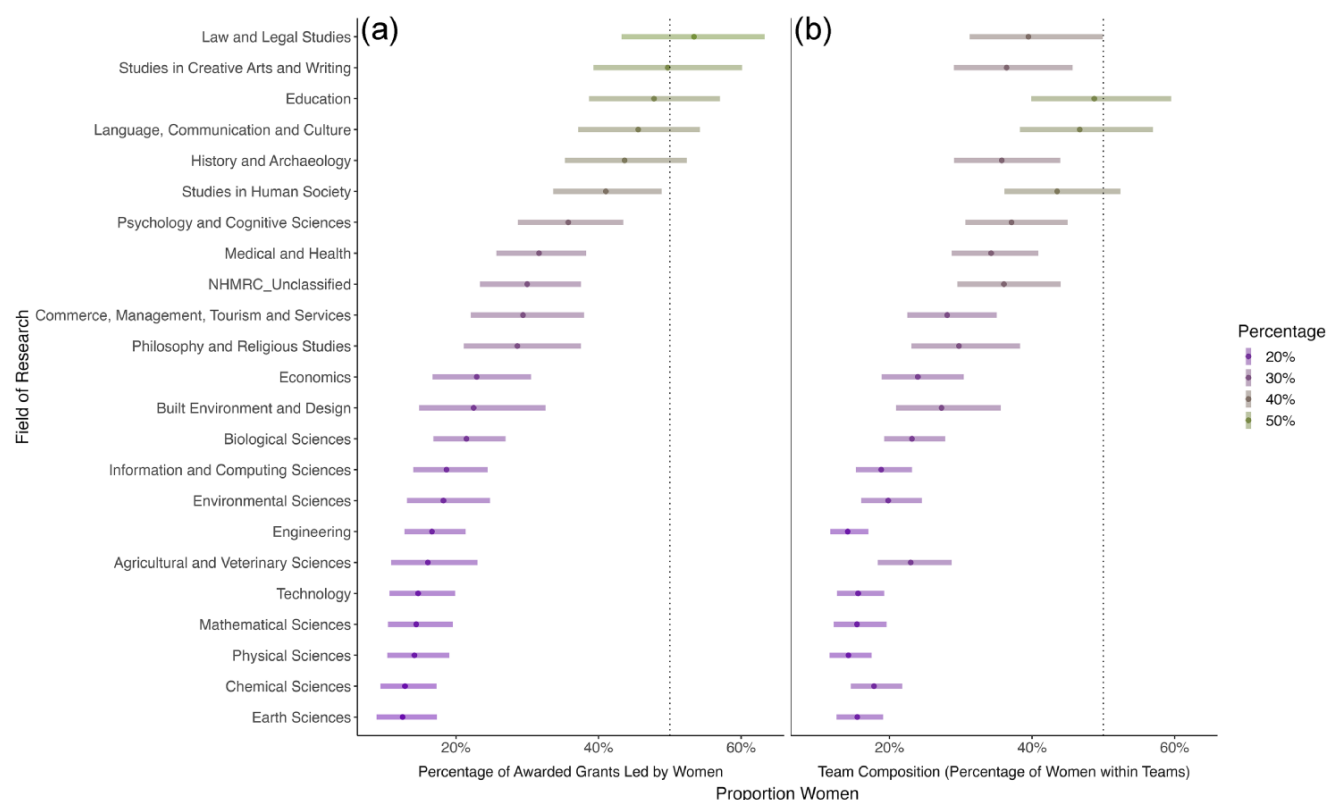


Figure 2. Awarded team grants led by women (RQ1) and within-team gender composition (RQ2) by field of research. Gender differences in awarded grants for both team grants and sole investigator grants varied by field of research (RQ1) (a). Within-team gender composition also varied by field of research (RQ2) (b). The dots represent the percentage of grants awarded to women relative to men for both team grants and sole investigator grants (a) or the average percentage of women within teams, excluding the principal investigator (b). The lines represent the upper and lower 95 per cent CI limits. For awarded grants (a), proportionally fewer awarded grants were led by women in the fields of earth sciences (13 per cent, CI 9, 17), chemical sciences (13 per cent, CI 9, 17), physical sciences (14 per cent, CI 10, 19), mathematical sciences (14 per cent, CI 10, 20), and technology (15 per cent, CI 11, 20). Five fields of research were consistent with gender parity, having proportions of awarded grants to women overlapping 50 per cent in their CIs: history and archaeology (44 per cent, CI 35, 52); language, communication, and culture (46 per cent, CI 37, 54); education (48 per cent, CI 39, 57); studies in creative arts and writing (50 per cent, CI 39, 60); and law and legal studies (53 per cent, CI 43, 63). These patterns did not differ between sole investigators and team leaders and mirrored the findings from the comprehensive analysis of the dataset (including grants from schemes exclusively catering to individual recipients) reported in Kingsley et al. (2023). Turning to within-team gender composition (b), grants awarded to teams of investigators fell short of gender parity (i.e. 50 per cent of women within a team) across all fields of research. Women were least represented among teams in the fields of physical sciences (14 per cent women within teams, CI 12, 17), engineering (14 per cent, CI 12, 17), mathematical sciences (15 per cent, CI 12, 20), earth sciences (15 per cent, CI 13, 19), technology (16 per cent, CI 13, 19), and chemical sciences (18 per cent, CI 15, 22). Four fields of research had within-team gender compositions consistent with gender parity: law and legal studies (39 per cent, CI 31, 50); studies in human society (44 per cent, CI 36, 52); language, communication, and culture (47 per cent, CI 38, 57); and education (49 per cent, CI 40, 60). These patterns did not differ by team leader gender. [Supplementary Table S3](#) presents the percentages of model estimates and the 95 per cent CI limits for each field of research.

engineering (14 per cent), mathematical sciences (15 per cent), earth sciences (15 per cent), technology (16 per cent), and chemical sciences (18 per cent). Four fields of research were consistent with gender parity, having gender compositions that overlapped 50 per cent of women within teams in their CIs (i.e. law and legal studies; studies in human society; language, communication, and culture; and education). These patterns did not differ by team leader gender (i.e. team leader gender was not found to be a significant source of variation with field of research in the secondary variable selection procedure).

4.3 Funding amounts by the gender and role of the principal investigator (RQ3) and by team gender composition (RQ4)

Funding amounts per grant did not differ according to the gender of the principal investigator. The effects of role and career

seniority of the principal investigator did not vary significantly by gender (RQ3), $F = 1.435$ (ns) and $F = 1.051$ (ns), respectively. Grants awarded to women team leaders received the same amount of funding per grant as men team leaders, relative to the scheme annual median, OR = 1.00 (CI 0.98, 1.02), as did grants awarded to women and men sole investigators, OR = 0.98 (CI 0.94, 1.02), when averaged across time, academic level, and research intensity of the host organization (Fig. 4). Among team leaders, women and men both received an average of 92 per cent (women: CI 85, 99; men: CI 86, 99) of the yearly scheme median.³ Among sole investigators, women received 80 per cent (CI 74, 87) and men received 82 per cent (CI 76, 89) of the yearly scheme median.

We observed a temporal trend towards funding parity across the principal investigator's gender over the period of 2000–20, $\beta = -0.01^{***}$, when averaged across academic level and research intensity of the host organization.

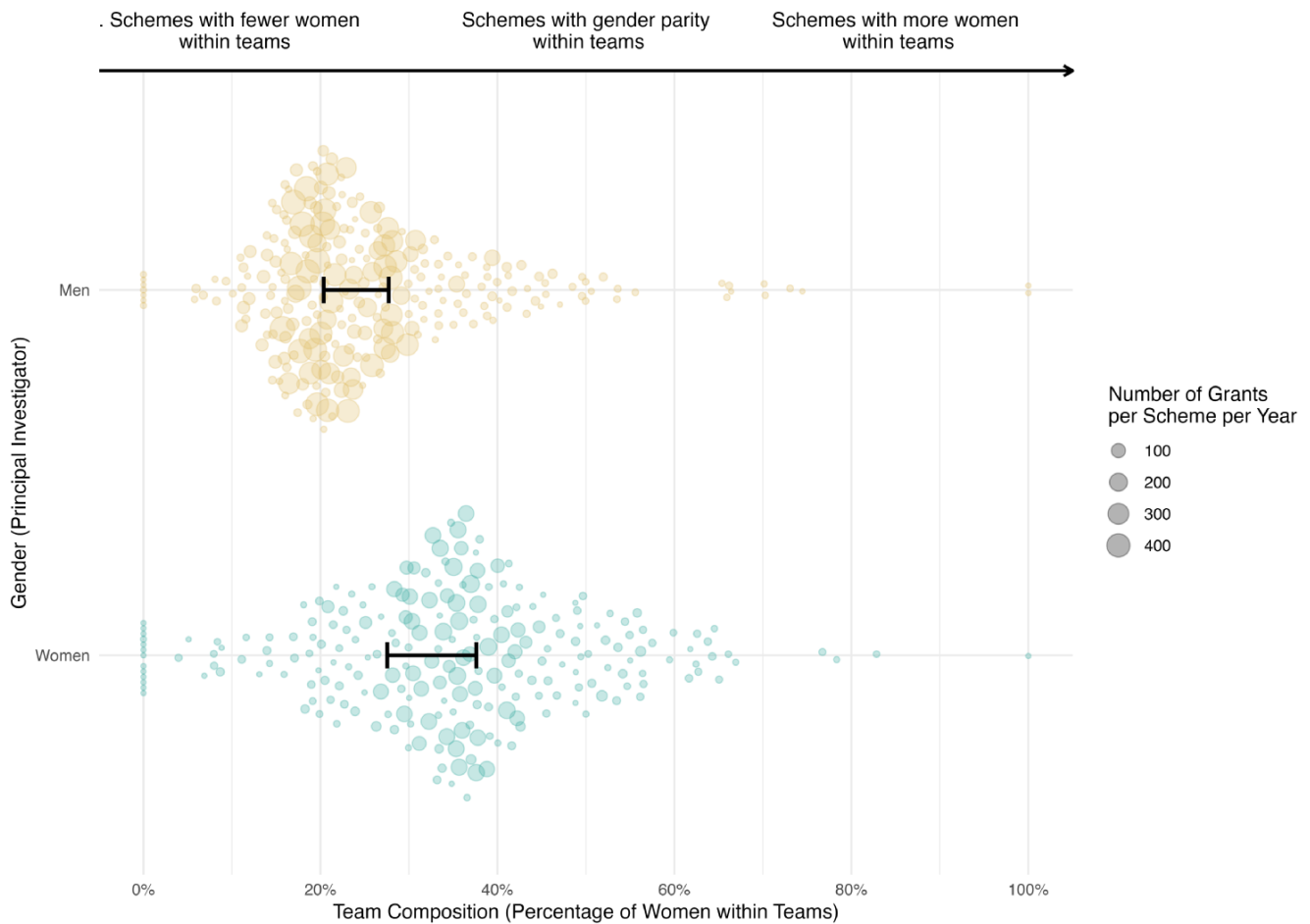


Figure 3. Percentage of women within teams led by women and men (RQ2). The average percentage of women within teams, excluding the principal investigator (y-axis), was higher for teams led by women (left) than for those led by men (right). These patterns did not vary across levels of career seniority of the principal investigator. The horizontal lines represent estimated marginal means, and the vertically widest points of the lines represent the upper and lower CI limits of those estimated means (95 per cent CI). The y-axis reflects average team composition per scheme per year. Each dot represents a scheme in a given year, and the size of the dots represents the number of grants in a given scheme and year. Dots are arranged in a beeswarm (Clarke, Sherrill-Mix, and Dawson 2017) to minimize overlap. Model estimates show that teams led by women were composed of 32 per cent women and teams led by men were composed of 24 per cent women, OR = **1.36** (CI 1.30, 1.41).

Turning to whether funding per grant differed by team gender composition (RQ4), we observed a trend over time towards funding parity across the 20-year period, $\hat{\beta}(\text{Team gender composition} \times \text{Time}) = 0.01^{***}$, when averaged across academic level and research intensity of the host organization. Historically, teams composed of mostly women (≥ 75 per cent women) received less funding per grant than teams composed of mostly men (≤ 25 per cent women). However, by 2020, funding amounts per grant were equivalent across all team gender compositions. Specifically, in 2000, teams composed of mostly women received on average 93 per cent of the yearly scheme median, while teams composed mostly of men received 100 per cent. By 2020, both majority and minority women teams received on average 85 per cent of the yearly scheme median.

Despite an overall pattern of equivalent funding amounts per grant within a scheme and year, there is a large disparity in cumulative funding amounts over two decades. Of the \$13.2 billion awarded to team leaders from 2000 to 2020, \$3.7 billion went to women-led projects and \$9.5 billion to men-led projects. Similarly, of the \$2.8 billion awarded to sole

investigators, \$0.7 billion went to women and \$2.1 billion to men. The accumulated difference across 20 years was \$7.2 billion less for women-led projects, \$5.8 billion less for women team leaders and \$1.4 billion less for women sole investigators. This outcome is driven by the numerically fewer awarded grants being led by women for both team grants and sole investigator grants.

5. Discussion

Women at mid- and senior-career levels were awarded a higher percentage of team grants than sole investigator grants compared to men at those levels. However, regardless of their role as team leader versus sole investigator, women were awarded fewer grants than men, although this varied by field of research. Teams led by women tended to include a greater percentage of women than men-led teams; however, overall, teams of investigators fell well short of gender parity regardless of the team leader's gender. Funding amounts per grant did not vary based on the gender of the principal investigator and, among team grants, reached parity across

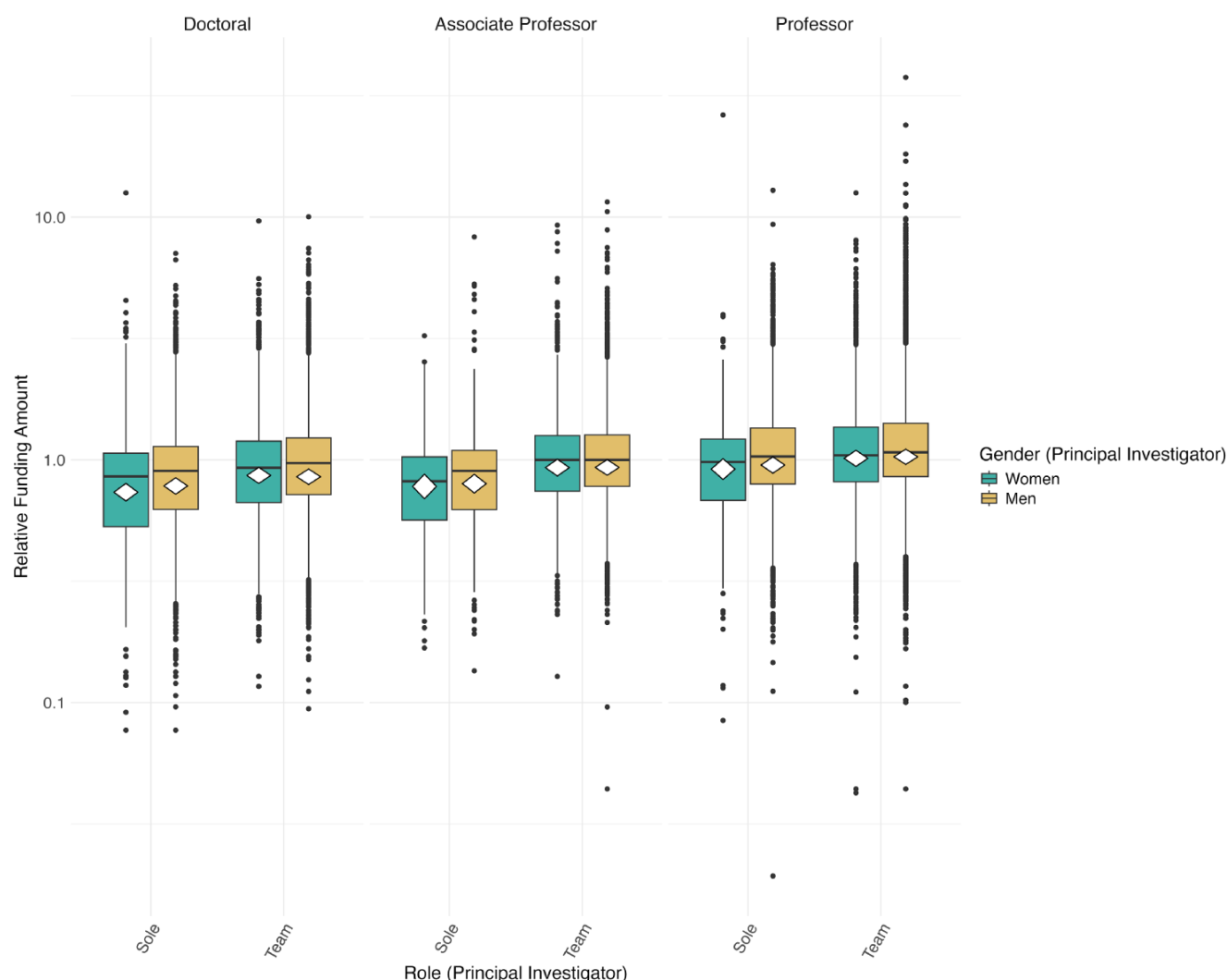


Figure 4. Funding amount relative to the scheme yearly median by the gender and role of the principal investigator (RQ3). Funding amounts per grant (y-axis) did not differ according to the gender of the principal investigator. Although team leaders were awarded more funding amounts per grant than sole investigators, and amount per grant increased with the principal investigator's career seniority, the effects of role and career seniority did not vary significantly by the gender of the principal investigator (RQ3). The boxplots illustrate the distribution of funding amounts per grant relative to the scheme yearly median separately for sole and team leaders and for women and men principal investigators. The horizontally widest part of the diamonds within the boxes represents the model estimated average funding amount relative to the scheme yearly median on the y-axis. Relative funding amounts by the gender, role, and academic level of the principal investigator, with a value of 1 indicating funding amounts per grant equivalent to the scheme yearly median and values greater (less) than 1 indicating higher (lower) funding amounts per grant relative to the scheme yearly median. Model estimates show that overall, within a scheme and year, grants awarded to women team leaders received the same amount of funding per grant as grants awarded to men team leaders, OR = 1.00 (CI 0.98, 1.02), as did grants to women and men sole investigators, OR = 0.98 (CI 0.94, 1.02), when averaged across time, academic level, and research intensity of the host organization. Funding amounts per grant to women relative to men did not vary according to the academic level. [Supplementary Figure S2](#) illustrates funding amounts per grant according to the gender of the principal investigator for NHMRC-awarded grants from 2000 to 2009 for which the academic title of the principal investigator was missing.

all team compositions in 2020. We note that our findings exclusively pertain to awarded grants, not application or success rates, and further research is needed to uncover insights across the entire grant lifecycle. These findings have implications for increasing women's representation in research grant programmes, with broader implications for promoting their retention and advancement in research.

Our methodological approach offered several strengths. We utilized a comprehensive dataset spanning two decades of Australian competitive government-funded research projects, providing a broad overview of gender differences, including

temporal trends. This study builds upon our previous comprehensive analysis of Australian research grant awards, applications, and workforce participation, which examined gender differences in awarded grants relative to application rates and research workforce participation by gender ([Kingsley et al. 2023](#)). While that study provided a broader context, the current analysis offers insight into collaboration patterns and team gender composition. We deployed statistical modelling that accounted for multiple explanatory factors enabling us to detect nuanced patterns of variation (e.g. based on gender, role, career seniority, and time) as well as make meaningful

inferences about gender differences after controlling for other known sources of variability. Importantly, our exploration of gender disparities considered the influence of career seniority and field of research, avoiding potential Simpson's paradoxes that have been documented in this area (Albers 2015; Volker and Steenbeek 2015).

Our study set out to answer four research questions (RQ1–RQ4), aiming to build on prior research and address gaps in the existing literature. Our findings advance the understanding of gender dynamics in the outcomes of competitive grant programmes, leveraging several strengths of our approach, with important implications for gender equity in research. First, our research offers an important advance by exploring collaboration patterns among principal investigators of awarded team research grants by gender, an area previously unexplored (RQ1). Our modelling points to women at mid- and senior-career levels being awarded a higher percentage of team grants than sole investigator grants compared to men at those levels. Second, our research builds on prior studies on gender composition of grant-funded research teams and gender homophily in research (RQ2). Consistent with prior research (Bowman and Ulm 2016; Pezzoni and Visentin 2024), we document that team gender composition of grants awarded to teams of investigators fell well short of gender parity, regardless of the gender of the team leader. Moreover, our findings reinforce the observation that teams led by women tend to have a higher proportion of women coinvestigators compared to teams led by men (Bowman and Ulm 2016; Pezzoni and Visentin 2024). This pattern is likely a manifestation of the well-documented phenomenon of gender homophily in research collaborations (McPherson, Smith-Lovin, and Cook 2003; Kwiek and Roszka 2021). Notably, our analysis highlights the importance of accounting for field of research when investigating gender dynamics. While some fields exhibited gender parity in the percentage of women within teams, others displayed a notable under-representation of women, echoing the field-specific variations documented in the comprehensive analysis of the full dataset (Kingsley *et al.* 2023). Another point of convergence with prior research is the evidence of a temporal trend towards gender parity in team composition across the study period (Pezzoni and Visentin 2024), particularly among teams led by early- and senior-career researchers. Together, our findings for research questions (RQ1 and RQ2) point to the potential of women's leadership of team grants in fostering gender equity in research grant programmes and beyond. Given the importance of research funding for career progression (Liner and Sewell 2009; Bonetta 2011), exploring avenues to accelerate the patterns observed in our research deserves attention. Doing so may also have broader implications for accelerating the attraction and retention of women in research (Kingsley *et al.* 2023).

Finally, our study makes a novel contribution by examining funding amounts, bringing clarity to the limited and mixed evidence on the influence of principal investigator role (RQ3) and team gender composition (RQ4) on grant outcomes. We document that funding amounts per grant did not differ by the gender of the principal investigator, and this pattern of gender parity did not differ between team leaders and sole investigators. While teams composed primarily of women historically received less funding per grant than teams composed primarily of men, this disparity diminished over time, reaching parity by 2020. Our findings to research

questions (RQ3 and RQ4) suggest an overall per-grant pattern of funding parity, despite a large cumulative funding disparity over two decades driven by numerically fewer grants awarded to women-led projects. Fostering women's leadership of team grants as an avenue to increase the representation of women in the research pool could, in turn, bridge the cumulative funding gap over time, extending the gender parity observed in individual grant amounts to the overall distribution of research funds. Together, our findings highlight the need to prioritize collaborative grant projects, support women's grant leadership, and incentivize team gender diversity to increase women's representation in research, providing a basis for evidence-informed action by research funders, governments, and higher education and research institutions.

Research funders and governments have social and legal responsibilities to ensure equitable opportunities for research funding for all researchers. They can implement gender-responsive funding mechanisms to promote diversity and collaboration, such as prioritizing collaborative grant applications and favouring gender-balanced research teams for proposals of the same score, as seen in Horizon Europe's ranking criteria (European Commission 2024). Mandating equal numbers of women- and men-led grant applications from a host institution and implementing gender quotas on grant awards can swiftly achieve gender equity (Weselingh 2024). Research funders and governments can also strengthen employer accountability for gender equity through eligibility requirements. Options include mandating gender equity plans (European Commission 2024; Snow Medical 2024), gender-based assessments (Canadian Institutes of Health Research 2022), or relevant accreditation as funding eligibility criteria—an approach with documented success in progressing gender equity (Ovseiko *et al.* 2020).

Higher education and research institutions can play a vital role in addressing structural inequities in the research workforce. Institutions can promote women's research leadership and incentivize collaborative efforts. Increasing women in supervisory and mentoring roles is effective in attracting and retaining women in research (Kewley *et al.* 2023). Incorporating collaboration criteria into performance evaluations can further encourage collaborative research efforts, independent of gender. Combining these strategies is likely to accelerate progress (Kewley 2021; Bergman *et al.* 2024).

To conclude, our findings reveal that women's team grant leadership may be an important mechanism for shifting overall representation of women researchers in the sector. This leadership could have broader implications in creating a positive ripple effect in the career pipeline, diversifying the research pool by attracting and retaining more women in research. Further research should examine the long-term impacts of this proposition on addressing gender equity in the research landscape.

6. Limitations

We wish to acknowledge limitations of this research. One limitation, mentioned earlier, is deploying a binary approach to gender, which is a nonbinary construct. With better data and reporting, future research will be able to explore potential biases involving nonbinary and gender diverse researchers. Our research also does not account for other documented sources of bias, such as ethnicity or cultural background

(Chen et al. 2022; Wild 2022; Nguyen et al. 2023), which can intersect with gender to compound disparities (Crenshaw 1989). Again, access to data on these constructs will accelerate the pace at which potential sources of bias, and their compounding impact, can be documented and attenuated. Another limitation, mentioned earlier, is that our dataset only included information on successful grant applications; therefore, our findings do not address application or success rates. While our previous comprehensive analysis (Kingsley et al. 2023) provides crucial context on these aspects, further research is needed to gain insights into the entire grant process. Finally, the nature of the available data precludes examination of differences in application quality, track records, or other contributing factors to research grant outcomes.

Acknowledgements

We thank the ARC and the NHMRC for their support and for providing data for this research that was not publicly available. We also thank Kate Nicholls, who performed desktop research to support this project, and Maeve McGillicuddy, who provided code to assist with data visualization.

Author contributions

Isabelle Kingsley (Conceptualization, Methodology, Investigation, Project administration, Formal analysis (descriptive statistics), Data curation, Writing—original draft), Eve Slavich (Conceptualization, Methodology, Investigation, Formal analysis (statistical models), Visualization, Writing—review & editing), Lisa Harvey-Smith (Conceptualization, Methodology, Writing—review & editing), Emma L. Johnston (Conceptualization, Methodology, Writing—review & editing), and Lisa A. Williams (Conceptualization, Methodology, Investigation, Writing—review & editing)

Supplementary data

Supplementary data is available at *SCIPOLE Journal* online.

Conflict of interest statement. I.K., L.H.-S., E.L.J., and L.A.W. received funding from the Australian Government Department of Industry, Science, and Resources that supported this research project via the Women in STEM Ambassador initiative. E.J. and L.A.W. also currently receive funding from the ARC. E.S. declares no competing interests.

Funding

This work was supported by a grant from the Commonwealth Department of Industry, Science and Resources (grant number WISA000001) that funded the Australian Government's Women in STEM Ambassador initiative, which was hosted by UNSW Sydney.

Data availability

Datasets required to reproduce the analyses are publicly available at <https://osf.io/2fr9t/>. The analysis code required to

reproduce the analyses and visualizations is publicly available at <https://osf.io/2fr9t/>.

Notes

1. In relation to the analysed data, we use the term 'women' for gender data classified as female and 'men' for gender data classified as male. We acknowledge the limitations of binarizing gender, as mentioned in this article's introduction.
2. In the model selection process, funding body (ARC versus NHMRC) and field of research were not significant unique predictors in any model.
3. To calculate estimated funding amounts for men and women, we calculated the weighted mean across years, weighting grants by the inverse of the count in their subgroup, where subgroups were the combination of gender and academic level of the principal investigator and field of research (e.g. principal investigator women professors in engineering). Across subgroups, 92%, 80%, and 82% represent the average funding amount relative to the scheme yearly median received in an average year. Unequally-sized subgroups (with professors being highly represented in the data) led to the unweighted scheme yearly median being relatively high and, thus, average funding amount percentages below 100%.

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